

AD 502 – Machine Learning

Pre-Reading Material – Lecture 25

Cure Algorithm

CURE Clustering Algorithm

Introduction

CURE stands for **Clustering Using REpresentatives**. It is a **hierarchical clustering algorithm** designed to handle large datasets and clusters of different shapes and sizes.

Many traditional clustering algorithms work well when clusters are roughly circular or similar in size. However, real-world datasets may contain clusters with irregular shapes, different sizes, and different densities.

CURE addresses this issue by representing each cluster using **multiple representative points** instead of using only a single central point.

Key Idea

CURE represents a cluster using several representative points that capture its shape and spread.

Why Do We Need CURE?

Consider a dataset containing customers, geographical locations, or image features. The data may form clusters that are:

- Irregular in shape
- Different in size
- Different in density
- Not necessarily centered around a single point

A method based only on the center of a cluster may not represent such groups accurately.



CURE uses several representative points to obtain a better description of the cluster.

Representative Points

The most important concept in CURE is the use of **representative points**.

Instead of selecting only one point to represent a cluster, CURE selects several points that are spread across the cluster.

These points help capture the **shape and structure** of the cluster.

For example, consider an irregularly shaped cluster. A single center point may not describe the cluster properly, while several representative points around the cluster can provide a better representation.

Shrinking Representative Points

CURE performs an additional step called **shrinking**.

After representative points are selected, they are moved slightly toward the center of the cluster.

This helps reduce the influence of:

- Noise
- Outliers
- Extreme data points

The resulting representative points provide a more stable representation of the cluster.

How CURE Works

The general working of CURE can be understood through the following steps:

Step 1: Select Data Points

CURE starts with the available dataset and considers the data points for clustering.

Step 2: Create Initial Clusters

Initially, individual data points or small groups are treated as separate clusters.

Step 3: Select Representative Points

For each cluster, several well-distributed representative points are selected.

Step 4: Shrink the Representative Points

The selected representative points are moved toward the cluster center to reduce the effect of outliers.

Step 5: Merge Similar Clusters

CURE compares representative points of different clusters. Clusters whose representative points are sufficiently close are merged.

Step 6: Continue the Process

The process continues until the desired number of clusters is obtained.

CURE and Outliers

One important advantage of CURE is its ability to handle **outliers** better than some traditional clustering approaches.

An outlier is a data point that is significantly different from the other observations.

Because CURE uses multiple representative points and shrinks them toward the center, extreme points have less influence on the final cluster representation.

CURE for Large Datasets

CURE was designed with large datasets in mind.

For very large datasets, processing every data point directly can require considerable computational resources.

CURE can use techniques such as **random sampling** to reduce the amount of data that needs to be processed.

The sampled data can then be clustered and used to help assign the remaining data points to appropriate clusters.

Advantages of CURE

CURE has several important advantages:

- Can identify clusters with irregular shapes
- Handles clusters of different sizes
- Uses multiple representative points
- More resistant to outliers than methods based on a single center
- Can be applied to large datasets
- Provides a hierarchical clustering structure
- Does not require all clusters to have the same shape

Limitations of CURE

CURE also has some limitations:

- The algorithm can be computationally expensive.
- The selection of representative points can affect clustering quality.
- The number of representative points needs to be chosen appropriately.
- Sampling very large datasets may affect the final clustering result.
- Choosing suitable parameters can require experimentation.

Applications of CURE

CURE can be applied to various real-world problems, including:

- Customer segmentation
- Geographic data analysis
- Image analysis
- Pattern recognition
- Market research
- Document clustering
- Biological data analysis
- Large-scale data mining

CURE vs. BIRCH

Both CURE and BIRCH are designed to handle large datasets, but their approaches are different.

Feature	CURE	BIRCH
Main idea	Multiple representative points	Compact cluster summaries
Cluster representation	Representative points	Clustering Features
Handles irregular shapes	Yes	More limited
Outlier handling	Relatively good	Depends on clustering structure
Large datasets	Suitable	Highly suitable
Structure	Hierarchical	CF Tree

Key Takeaway

CURE is a **hierarchical clustering algorithm** that represents each cluster using **multiple representative points**. These points are selected to capture the shape and spread of the cluster and are then moved toward the cluster center to reduce the effect of outliers.



CURE = Multiple representative points + shrinking + hierarchical clustering

Pre-reading focus: Students should understand **why multiple representative points are used, how representative points are shrunk, how clusters are merged, and why CURE is useful for irregularly shaped clusters.**

